Now that you have a better understanding of the role middleware plays, you should understand how it is implemented, so you can provide your own custom functionality to the request pipeline. This section will cover creating your own middleware components and registering them with the pipeline.

Middleware components are implemented as `RequestDelegate`s—that is, a .NET delegate that takes an `HttpContext` as a parameter and returns a `Task`, or, in other words, a method that represents an asynchronous operation on the `HttpContext`:

```
public delegate Task RequestDelegate(HttpContext context)
```

Delegates 101: a quick refresher

Delegates in .NET are types that represent a method signature and return type. The following example declares a delegate named `MyDelegate` that takes a `DateTime` as a parameter and returns an integer:

```
delegate int MyDelegate(DateTime dt);
```

Any method that has the same signature and return type can be assigned to an instance of `MyDelegate` and invoked, including both of the methods shown next.

Assigning methods to delegates based on matching signatures and returning types

```
int GetMonth(DateTime dt)                    ❶
{
 return dt.Month;
}
int PointlessAddition(DateTime dt)           ❶
{
    return dt.Year + dt.Month + dt.Day;
}
```